RESEARCH + SOFTWARE

ZHONGJIE HUANG

AI & Computer Science

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PROFILE

I work across computer vision, machine learning evaluation and software development. My projects connect model experiments with usable systems, with particular attention to what the evidence supports and where it stops.

EDUCATION

University of Sheffield

2025-2026

MSc Artificial Intelligence

Dissertation submitted; final results and degree award pending.

University of Sheffield

September 2021 - July 2025

BSc Artificial Intelligence and Computer Science

Awarded 1 July 2025 • Upper Second Class (2:1)

SELECTED ACADEMIC PROJECTS

Cell segmentation & downstream evidence

2026

Individual MSc dissertation project

- Compared QuPath, Cellpose-DINO, Cyto R-CNN and CSGO pipelines, examined their quantitative outputs, and assessed downstream models with attention to grouping and the independence of evaluation data.
- The dissertation did not establish a clear gain from fusion over the image-only approach. This makes the distinction between a better segmentation output, more useful measurements and better predictions central to the interpretation.

Python / QuPath / Cellpose-DINO / Cyto R-CNN / CSGO / Grouped evaluation

Project summary

Emotion recognition using visual information

2025

Individual undergraduate project

- A complete image-to-prediction workflow, linking a Swin Transformer model with a responsive Java/Tomcat web application.
- The final report records 76.25% accuracy on its mixed test set.

Python / PyTorch / Swin Transformer / Java / Tomcat / HTML / CSS / JavaScript

Project summary

PROPOSED DOCTORAL DIRECTION

Reliable visual segmentation under environmental and sensor shift

A proposed doctoral direction: investigate whether confidence calibration, feature novelty and perturbation sensitivity help identify unreliable segmentation predictions when data conditions change. These are future research questions, not completed experiments.

SELECTED SOFTWARE PROJECTS

Marginloom - AI reliability workbench

2026

Personal research and software prototype, developed with AI assistance

- The repository implements temperature scaling, risk-coverage and threshold exploration, validated JSON prediction imports, saved run and policy snapshots, local ensemble inference, persisted review notes and JSON evidence exports. A separate companion experiment demonstrates segmentation on synthetic images.
- A React and TypeScript interface connects to typed API handlers and SQLite/D1 persistence. Python and scikit-learn produce reproducible baseline models and evaluation exports.

React / TypeScript / Python / scikit-learn / SQLite / D1 / Docker

Project summary | Public source

IAFM - Investment & financial management

Portfolio project

Personal software project

- A Java MVC application for multi-currency financial reports, paired account transfers and administrator-managed inflation records.
- The existing Java MVC structure separates request handling, domain logic and persistence. Server-side account permissions control inflation maintenance, while transfer entries are written and reversed together in database transactions.

Java / SQL / Domain modelling / Access control / Dashboard UX

Project summary | Public source

TECHNICAL TOOLKIT

Programming & interfaces: Python, Java, TypeScript, React, HTML / CSS / JavaScript

Machine learning & vision: PyTorch, Swin Transformer, OpenCV, QuPath, Cellpose-DINO, Cyto R-CNN, CSGO, scikit-learn

Data & engineering: NumPy, pandas, Matplotlib, SQL, SQLite / D1, Git, Docker, Linux, Tomcat, GitHub Actions

Research practice: Baselines, grouping and leakage checks, error analysis, traceable experiments, literature synthesis and clear statements of limitations.

Communication: Chinese; English - used in UK university study and technical communication